Homework 4

In this homework, you will work with Sonic Visualizer and Sonic Annotator tools. You received the following files with the homework:

Homework file	What is in it
Instruction.pdf	The instruction you are reading now
Audio pack	Folder containing 5 .mp3 files from Internet Archive
Transforms	Example transform for Sonic annotator
Report files	A folder for the report that you need to zip up and upload back to Moodle

Setup

Sonic Annotator is the command-line companion to Sonic Visualiser. Presumably, you have already installed it during the practice. If not, download it and note the path to it, which we will refer to as PATH_TO_SA.

A guided tour of Sonic Annotator

During the practice, we learned to use Sonic Visualizer and plot different features with it. Now, we are going to learn to write them into a file using Sonic Annotator.

What plugins do I have?

Open a command line and write this command:

```
$ sonic-annotator -l
```

-l lists every plugin output that Sonic Annotator can see on your system.

Output is a long list of identifiers of the form vamp:<library>:<plugin>:<output>. For example:

```
vamp:qm-vamp-plugins:qm-mfcc:coefficients

vamp:vamp-example-plugins:spectralcentroid:linearcentroid

vamp:nnls-chroma:nnls-chroma:chroma

vamp:vamp-libxtract:spectral_inharmonicity:spectral_inharmonicity
```

Each identifier names exactly one output of one plugin. Many plugins have multiple outputs — for example the QM Onset Detector extracts onsets, detection_fn and smoothed_df — and you have to specify which one you want.

Generate a starting transform file

A **transform file** (.n3) is a tiny RDF document that says "run this plugin output, with these parameters". You don't need to write it yourself. You ask Sonic Annotator for a skeleton with -s, then edit it.

```
$ sonic-annotator -s vamp:vamp-example-plugins:spectralcentroid:linearcentroid > centroid.n3
```

Open the file and look at it. What are the parameters that you can change here? Three things, actually:

- **block_size** — the FFT window length in samples. Bigger = finer frequency resolution, coarser time resolution. The window size should also depend on the sampling frequency. There is a default given, assuming a sampling frequency of 44100 Hz.
- **step_size** — the hop between successive FFT frames. Smaller = denser output. Conventionally 1/2 or 1/4 of block_size.

> **Do I need a transform .n3 file?** There is also a shortcut -d flag that uses all the defaults. However, defaults can change between releases. A saved .n3 is reproducible and can be adapted to your situation.

Run a single transform on one file. Copy the following line into your command line:

```
sonic-annotator -t centroid.n3   -w csv --csv-stdout audio_pack/VeniCreatorSpiritus-CantoGregoriano.mp3
```

--csv-stdout dumps the result to your terminal so you can sanity-check it. Drop that flag once you trust the output and CSVs will be written to disk instead.

The whole directory, all at once

Now try to point Sonic Annotator at the folder with -r (recursive), set an output directory with --csv-basedir, and tell it to overwrite existing CSVs with --csv-force:

```
$ mkdir features

$ sonic-annotator -r  -t centroid.n3 -w csv   --csv-basedir features  --csv-force audio_pack/
```

Sonic Annotator will produce one CSV per (audio file × transform). Filenames follow the pattern <audioname>_vamp_<library>_<plugin>_<output>.csv (with colons in the plugin URI replaced by underscores).

Multiple transforms in one run

You can pass -t as many times as you want: Sonic Annotator runs them all on each audio file in one pass:

```
$ sonic-annotator -r
  -t transforms/centroid.n3
  -t transforms/mfcc.n3
  -t transforms/onsets.n3
  -w csv --csv-basedir features --csv-force
  audio_pack/
```

Summary statistics (the optional fast path)

Sonic Annotator can collapse a per-frame feature into a single number per file, with the -S flag. Available summaries: min, max, mean, median, mode, sum, variance, sd, count. Combine with --summary-only to skip the per-frame output entirely:

```
$ sonic-annotator -r -t centroid.n3 -S mean -S sd --summary-only -w csv --csv-basedir features --csv-force audio_pack/
```

CSV output format

Sonic Annotator's CSVs have **no header row**. The columns are, in order:

Column	Meaning
1	**time (seconds)** — start time of the feature frame
2	**duration (seconds)**, if the plugin produces features with duration; otherwise omitted
3, 4, …	**feature values** — one column per bin. A scalar feature (centroid) uses one column; MFCCs use 13; a chroma vector uses 12.
last	**label** if the plugin emits one (e.g. chord names from Chordino)

Your homework task

Explore the audio files in the folder using Sonic Visualizer. Compute different audio features on these three files and examine them visually. Then, pick **three features** of your choice.

When picking, think about which feature would actually *say something* about this corpus. The five recordings differ in era, instrumentation, recording quality, and texture. A feature that picks up on any of those is interesting.

Here are some suggestions:

Family	Plugin output	What it measures	Good musical question
Timbre	vamp:vamp-libxtract:spectral_centroid:spectral_centroid	Centre of mass of the spectrum in Hz. High = bright/harsh, low = dull/dark.	Which recording is the brightest?
Timbre	vamp-libxtract: spectral_inharmonicity: spectral_inharmonicity	Fraction of spectral energy outside the predicted harmonic series.	Distorted brass, cymbals, and noise raise this.
Rhythm	vamp:qm-vamp-plugins:qm-onsetdetector:onsets	Note onset times.	Mean onsets per second per track.
Rhythm	vamp:bbc-vamp-plugins:bbc-rhythm:rhythm-strength	How periodic the onset structure is.	Which recording has the strongest pulse?

Harmony	vamp:nnls-chroma:chordino:chordnotes	Chord labels in time	How much harmonic change is there?

1. **Extract the features of your choice and attach the extracted features as csv.**
2. **Defend your choice – why did you pick these features? What do they say about audio files? Here you can attach screenshots from sonic visualizer to show how these features differ between these files and why. Write 100-200 words about your choices, your hyperparameter choices and how they relate to the music you analyze.**

Bonus exercise (3 points)

Upload you CSV files to a colab computer, and work with AI to analyze your results statistically (create histograms, boxplots and other types of graphs suitable for your features).